

My thought process for what I chose:

I decided to go with the budget tracker project because out of all the options it felt like the one that most naturally covers everything we learned in chapters 8 through 11 without me having to force anything in just to check a box. The basic idea is that the program lets a user track their money by recording income and expenses, sorting them into categories, and then saving all of that data so it is still there the next time they open the program. For the object oriented part I am using a base Transaction class that holds the shared information like the amount, the date, and the category, and then two subclasses called Income and Expense that inherit from it. This is where the inheritance comes in, because an income and an expense really are two specific types of the same general thing, but they behave a little differently, like how income adds to the balance while an expense subtracts from it. On top of that I have a Budget class that holds all the transactions and has methods to add them, remove them, and calculate totals and category breakdowns, which is composition since the budget has transactions rather than being one. I am splitting the code into separate modules instead of one big file, so the classes, the saving and loading logic, and the main program that runs everything are all kept in their own places. For saving the data I am using the json module so the whole budget can be written to a file and loaded back in, and I am wrapping the file handling in try except blocks so the program does not crash if the file is missing on the first run or if the data gets messed up. Finally I am writing unit tests to make sure the core logic actually works, like checking that adding transactions gives the right balance and that the category totals come out correct, which covers the chapter 11 testing requirement.